Lab Report – 09

Course Code: CSE 412

Course Title: Computer Graphics Lab

Experiment Name: Two-dimensional Object Animation

Submitted To:

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Two-dimensional Object Animation

Title: Multiple Animated Objects Using OpenGL

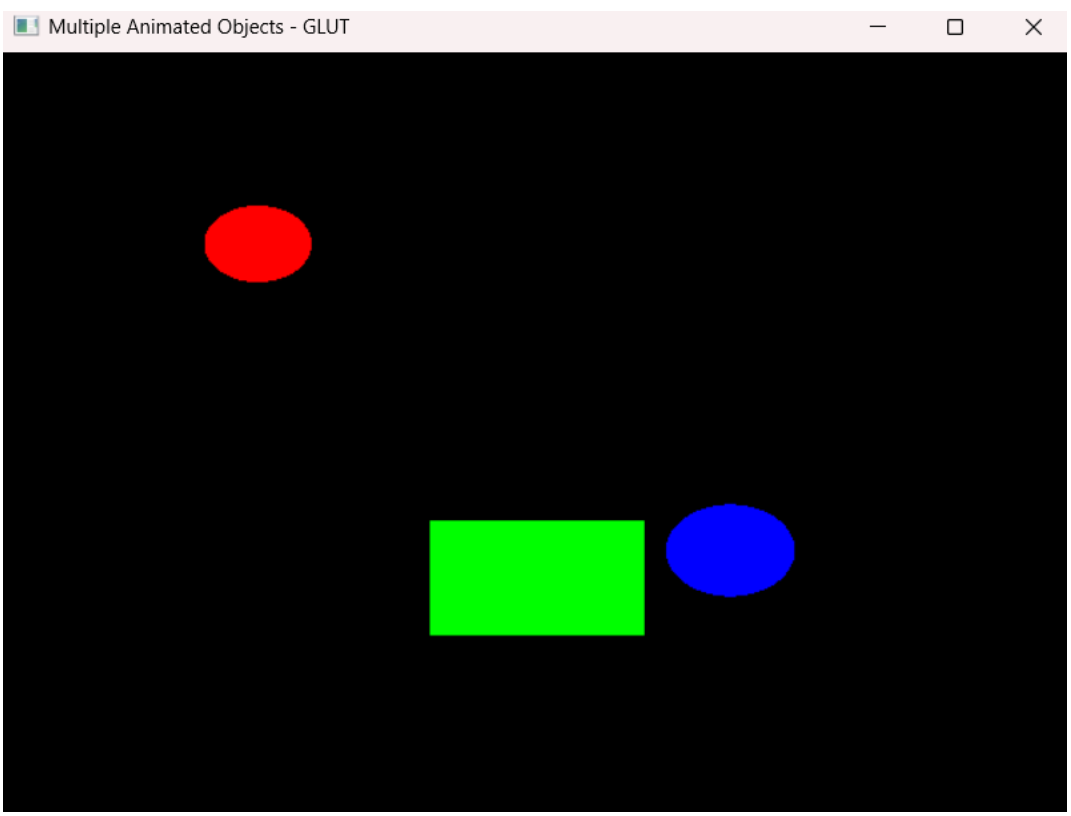
Introduction:

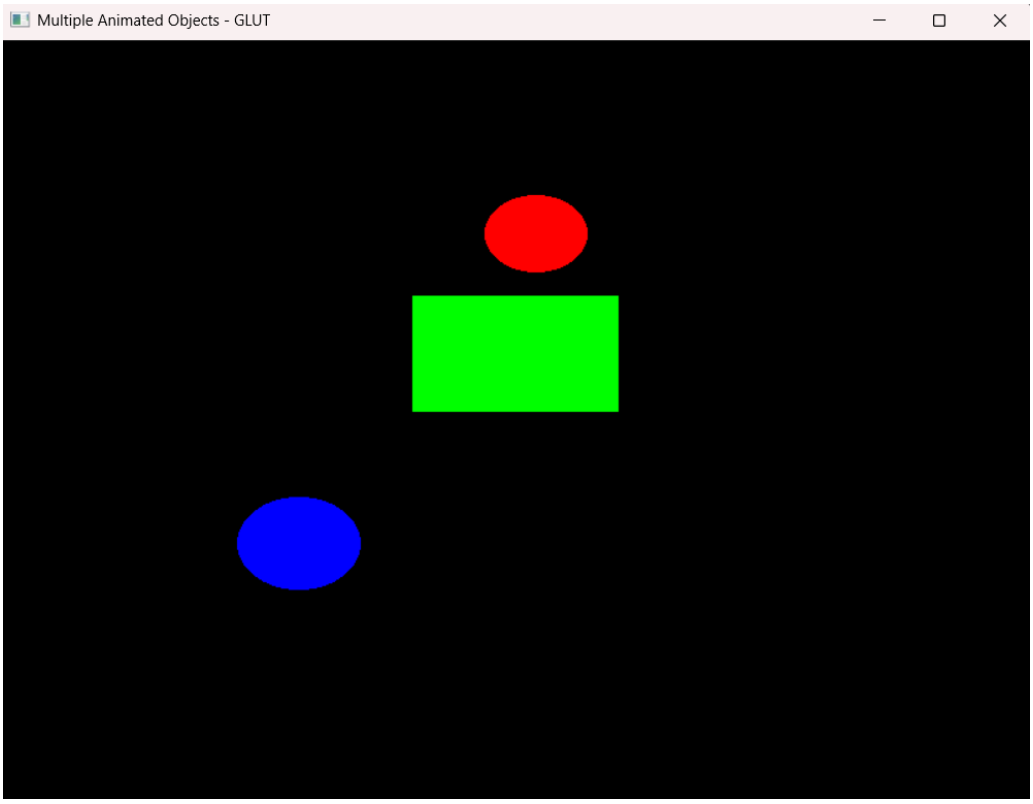
In this project, multiple shapes like circles and a square are drawn using OpenGL. These objects are animated and move continuously in different directions.

Contents:

- **Functions Used:** `drawCircle(), glBegin(), glVertex2f(), glutTimerFunc()`
- **Concept Used:** Animation (Movement using Timer)
- **Shape Used:** Circle and Square

Graph:





Draw two circles moving left and right, and one square moving upward continuously.

Code (Drawing):

```
// display function
void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    // Moving Circle 1
    glColor3f(1, 0, 0);
    drawCircle(x1, 0.5f, 0.1f);

    // Moving Circle 2
    glColor3f(0, 0, 1);
    drawCircle(x2, -0.3f, 0.12f);
```

```
    // Moving Square
    glColor3f(0, 1, 0);
    glBegin(GL_QUADS);
        glVertex2f(-0.2f, y_pos);
        glVertex2f( 0.2f, y_pos);
        glVertex2f( 0.2f, y_pos + 0.3f);
        glVertex2f(-0.2f, y_pos + 0.3f);
    glEnd();

    glFlush();
}

// animation timer
void timer(int) {
    x1 += 0.01f;
    x2 -= 0.015f;
    y_pos += 0.01f;

    // reset positions
    if (x1 > 1.2f) x1 = -1.2f;
    if (x2 < -1.2f) x2 = 1.2f;
    if (y_pos > 1.2f) y_pos = -1.2f;

    glutPostRedisplay();
    glutTimerFunc(16, timer, 0);  // ~60 FPS
}
```

Source Code: GitHub

Discussion:

The task was to draw multiple objects and animate them. I used timer function to move the shapes continuously in different directions.